



233/3
CHEMISTRY PAPER 3
MARKING SCHEME

QUESTION 1

PROCEDURE 1

TABLE 1

	i.	ii.	iii.
Final burette reading (cm ³)	20.0	20.0	20.0
Initial burette reading (cm ³)	0.0	0.0	0.0
Volume of solution c used (cm ³)	20.0	20.0	20.0

C.T -1
A.P -1
AC-1
AV-1
F.A-1
5

- a)
- Average volume = $\frac{20.0 + 20.0 + 20.0}{3} = 20.0 \text{ cm}^3$
 - Moles of solution C = $\frac{0.125 \times \text{titre} \sqrt{1/2}}{1000} = \text{Ans} \sqrt{1/2}$
 - Moles of solution D = $1 \times \text{ans a(ii)} = \text{Ans} \sqrt{1/2}$
 - Molarity of solution D = $\frac{\text{Ans a(iii)} \times 1000 \sqrt{1/2}}{25} = \text{Ans} \sqrt{1/2}$
 - Molarity of solution B
 $M_{\text{conc.}} \times V_{\text{conc.}} = M_{\text{di}} \times M_{\text{di}} \sqrt{1/2}$
Molarity = $\frac{\text{Ans a(iv)} \times 150 \sqrt{1/2}}{50} = \text{Ans} \sqrt{1/2}$

b) Table 2

	i.	ii.	iii.
Final burette reading (cm ³)	15.8	15.8	15.8
Initial burette reading (cm ³)	0.0	0.0	0.0
Volume of solution c used (cm ³)	15.8	15.8	15.8

C.T-1
D.P- 1
AC-1
AV-1
F.A-1
05

- b)
- Average volume of sol A = $\frac{15.8 + 15.8 + 15.8}{3} = 15.8 (\text{cm}^3)$
 - Moles of solution D used = $\frac{\text{Ans a(iv)} \times 25 \sqrt{1/2}}{1000} = \text{Ans} (\text{cm}^3) \sqrt{1/2}$
 - Moles of sol A = $\frac{\text{Ans b(ii)} \sqrt{1/2}}{2} = \text{Ans} \sqrt{1/2}$
 - Solubility of solid A
Titre volume = Ans b (iii)
 $100 \text{ cm}^3 = \frac{100 \times \text{Ans b (iii)}}{\text{Titre}} = \text{moles} \sqrt{1/2}$
Solubility = $\frac{100 \times \text{Ans b (iii)} \sqrt{1/2}}{\text{Titre}} \times 126$
= $\text{Ans} \sqrt{1/2}$

2.

No.	Observation	Inference
a)	Colour less filtrate $\sqrt{1/2}$ white residue $\sqrt{1/2}$	$\text{Cu}^{2+}, \text{Fe}^{2+}, \text{Fe}^{3+}, \sqrt{1/2}$ absent in both filtrate and residue $\sqrt{1/2}$
I.	White ppt $\sqrt{1/2}$ soluble in excess $\sqrt{1/2}$	$\text{Zn}^{2+}, \text{Pb}^{2+}, \text{Al}^{3+}, \sqrt{1}$
II.	White ppt $\sqrt{1/2}$ soluble in excess $\sqrt{1/2}$	$\text{Zn}^{2+} \sqrt{1}$
III.	White ppt $\sqrt{1/2}$	$\text{SO}_3^{2-} \sqrt{1/2} \quad \text{CO}_3^{2-} \quad \text{Cl}^-$
	Effervescence / bubbles of colours less odour less gas $\sqrt{1/2}$	$\text{CO}_3^{2-} \sqrt{1/2}$
	White ppt $\sqrt{1/2}$ soluble in excess $\sqrt{1/2}$	$\text{Pb}^{2+}, \text{Zn}^{2+}, \text{Al}^{3+} \sqrt{1}$
	White ppt $\sqrt{1/2}$ insoluble in excess $\sqrt{1/2}$	$\text{Pb}^{2+}, \text{Al}^{3+} \sqrt{1}$
	White ppt formed	$\text{Pb}^{2+} \sqrt{1/2}$



3.

No.	Observation	Inference
a)	Solid melts and burns with yellow sooty flame ✓1	$\begin{array}{c} \diagdown \\ \text{C} = \text{C} \diagup \end{array} \quad \text{✓1} \quad \text{C} = \text{C} -$
b)	Purple potassium manganate Solution turns colourless ✓1	$\begin{array}{c} \diagdown \\ \text{C} = \text{C} \diagup \end{array} - \text{C} \equiv \text{C} - \text{✓1} \quad \text{R-OH}$
	Effervescence / bubble of colourless gas ✓1	$\begin{array}{c} \text{O} \\ \parallel \\ \text{H}^+ / -\text{C} - \text{OH} \end{array} \text{✓1}$